COURSE: Database Fundamentals (DBF510S)

LAB 3: Conceptual Database Design

Due Date: 22nd March 2025

NB: Exercise 1 and 2 to be completed in pairs during practical class.

Students should use drawio https://app.diagrams.net/ to complete exercise 3 and submit on eLearning. The ER diagram can be saved as a PDF and submitted on elearning.

E-R diagram using Crows Foot Notation

Exercise 1. Given the following business rules. Identify all the entities. Choose at least 5 attributes for each entity and include a primary key.

a) A Student can enroll in many Courses,
b) A Course can have many Students enrolled
c) A Course can have multiple projects
d) A project can have multiple student working on it.

Exercise 2. Given the following business rules. Identify all the entities. Choose at least 4 attributes for each entity and include a primary key.

a) A student can have one or two parents
b) Parent can have one or more students
c) A student can have multiple siblings
d) A sibling can be related back to several students

Exercise 3. Given the following business rules. Identify all the entities. Choose at least 4 attributes for each entity and include a primary key. (Total Marks 30)

a) A pensioner can enroll in one pension plan
b) A pension plan can have many pensioners
c) A pensioner can receive multiple payments
d) Each payment is for one specific pensioner
e) A pensioner can have multiple dependents
f) Each dependent is associated with only one pensioner